

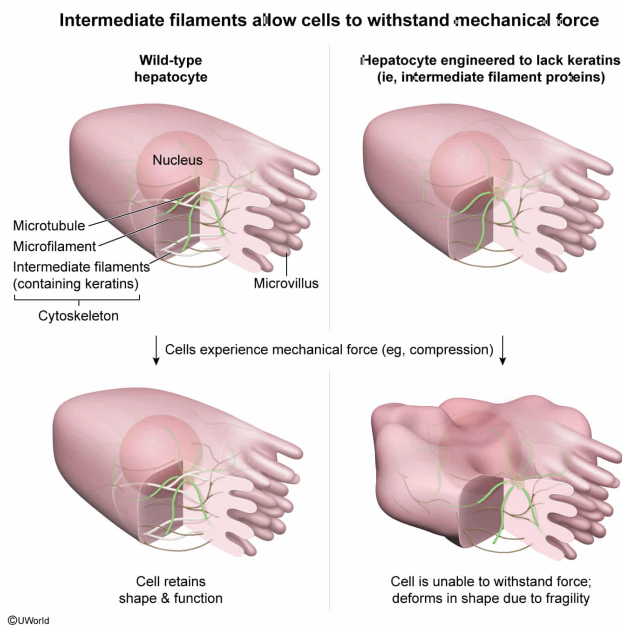
Keratins are a large family of intermediate filament proteins. Scientists studying the effect of keratins in the liver generated model organisms deficient in keratins. These organisms may display abnormal hepatocyte phenotypes because keratin deficiency may reduce the:

- ☐ A. ability of cells to undergo cytokinesis. [10%]
- ☐ B. intracellular transport of substances. [15%]
- ✓ ☒ C. capacity of cells to resist mechanical forces. [66%]
- ☐ D. number of cilia on the cell surface. [7%]

Correct

65% answered correctly

Explanation:



Eukaryotic cells contain an internal **cytoskeleton**, which acts as a **framework** that provides shape and internal structure to cells and helps organize intracellular components. The cytoskeleton is made of a network of fibers interspersed throughout the cytoplasm and consists of **three major components**: **intermediate filaments**, **microfilaments**, and **microtubules**.

Intermediate filaments are composed of several protein types (eg, keratin, lamin) and perform the following functions:

- Work with microfilaments to determine cellular shape
- Make up the nuclear lamina (inner lining of nuclear envelope)
- Assist in anchoring **organelles** to specific compartments within the cell

- **Provide crucial support for the cell to be able to withstand mechanical forces (eg, compression)**

In this question, scientists are studying the effect of keratins (a large family of *intermediate filament* proteins) in hepatocytes ([liver cells](#)). They generated model organisms deficient in keratins, which would make the organisms' **hepatocytes** more fragile and **unable to withstand/resist mechanical forces**. Accordingly, these model organisms may display abnormal hepatocyte phenotypes.

(Choice A) Microfilaments are made of actin protein subunits organized into double-stranded rods. Microfilaments help determine the overall shape of a cell and assist in some cellular locomotion. They also are responsible for [muscular contractions](#) and forming the [cleavage furrow](#) during cytokinesis. Therefore, reduced ability of a cell to undergo cytokinesis would be a result of the cell's being deficient in microfilaments, *not* intermediate filaments.

(Choice B) Microtubules are made of alternating α - and β -tubulin subunits organized into hollow tubes. Microtubules form the mitotic spindle during the [cell division process](#). They also are involved in movement within the cell; for example, they facilitate intracellular [vesicular transport](#). Therefore, reduced intracellular transport of substances would be a result of a deficiency in microtubules, *not* intermediate filaments, in the cell.

(Choice D) Microtubules form the cores of both [cilia](#) and [flagella](#). A cell having a reduced number of cilia on its surface would be a result of a microtubule, *not* intermediate filament, deficiency.

Educational objective:

The cytoskeleton consists of three types of fibers: microfilaments, microtubules, and intermediate filaments. Intermediate filaments are important in determining cell shape, anchoring organelles, and allowing cells to withstand mechanical stresses.

Time spent:0 sec
Subject:
Biology

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System:
Digestion and Excretion